TRAJECTORYLAB

User Manual

Scientific environment for modelling the motion of spherical objects

ENGLISH EDITION

- Document version: TrajectoryLab 0.4.2
- Interface languages: Russian and English
- Platform: Windows 10/11 x64
- Stack: Python · PySide6 · Qt 6 · pyqtgraph · Pint · SciPy
- Scope: low-speed laboratory and sports experiments
- Model: point mass / rigid sphere, translational motion
- Prepared for local scientific use

This manual describes features provided by the source project. Actual behaviour depends on the installed build and its version.

TrajectoryLab · pomidorkmen

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Purpose and scientific scope

TrajectoryLab is a local scientific environment for reproducible experiments involving translational motion of a rigid spherical particle. It combines atmosphere, drag, gravity and numerical integration.

The baseline model does not describe orientation, spin, lift, droplet deformation, collisions, wall effects or complex flow geometry. Such tasks require a dedicated model or CFD.

Use the software for educational, laboratory and sports experiments involving low-speed spherical objects. Always consider the Cd(Re) validity range, medium properties and numerical error.

Control / action	Description
Suitable	Rigid sphere; translational motion.
Not included	Spin, lift, deformation and particle interactions.
Verify	Geometry, flow regime and Re range.

Practical note
Agreement with one measured path does not prove a model is universal. Test sensitivity to step size, atmosphere and Cd.

Installation and first launch

Extract the complete archive into a writable folder. build.exe, build.py, pyproject.toml and project directories must remain together. Do not run the executable from inside the ZIP viewer.

Double-click build.exe. On first launch it may download the official Python installer from python.org, verify the pinned SHA-256 and install a private runtime under the current user's profile.

Builder creates a virtual environment, installs dependencies with pip, checks package consistency, runs pytest and source checks, then packages the Qt application with PyInstaller.

After packaging, choose the initial theme in Builder. A shortcut is created in the actual Windows Desktop folder. Administrator privileges and system PATH changes are not required.

The first setup needs internet access to python.org and PyPI. Ordinary calculations in the packaged application are local and do not require internet.

Control / action	Description
Platform	Windows 10/11 x64.
Permissions	Write access to the current user's profile.
First run	Internet is needed for the runtime and packages.

Practical note
A SmartScreen unknown-publisher warning is not the same as a malware detection. Do not override a specific threat alert.

Builder: main window

Builder uses a compact scientific-utility layout: light grey panels, thin borders, system UI font and a monospaced journal. Main build stages remain visible in one work window.

The RU and EN flag buttons at the top change labels, groups, statuses and journal language. The version label sits quietly next to the application title.

Build options include tests, cleanup of previous outputs and onedir/onefile mode. Use onedir for normal distribution and do not skip tests for a release.

The progress bar shows the current stage; the journal contains commands and output from pip, pytest and PyInstaller. On failure, save the complete journal, not only its last line.

After packaging, Light/Dark/Soft Pink selects the application's initial theme. This stores a setting; it does not recompile the scientific core.

Control / action	Description
Language	Flags switch labels and messages.
Progress	Stages: environment, tests, PyInstaller, verification.
Result	Paths to executable, archive and shortcut.

Practical note

Do not close Builder during installation and packaging. On network errors, inspect status and journal first rather than disabling Windows protection.

First application start

Launch the packaged TrajectoryLab by double-clicking its EXE or shortcut. The main window restores saved language, theme and dock layout. A demonstration setup is loaded on first start.

Click the run button in the left dock or press F5. Before a run, the interface validates dimensions, allowed ranges and model configuration.

After the run, inspect the results dock and central tabs. If OpenGL is unavailable, use 2D, time-series and table views; the physics core runs independently from 3D visualization.

Open the vacuum example and verify its analytical solution. Then run the air-resistance example and inspect speed and Reynolds behaviour.

The application does not need Administrator privileges. If the shortcut is missing, run the EXE directly from dist and inspect the Builder journal.

Control / action	Description
Trial run	F5 or the RUN button.
Check	Vacuum example before drag models.
3D	Missing OpenGL does not imply a calculation error.

Practical note
For the first check, do not change the demonstration inputs; first confirm that the baseline installation works.

Main window: layout

The main window is a QMainWindow: menus and toolbar at the top, parameters on the left, results on the right, journal at the bottom, and plot/table tabs in the centre.

File contains open/save, JSON/CSV/PDF export and exit. Experiment starts a run and opens the model manager.

View controls three dock panels, RU/EN language and Light/Dark/Soft Pink theme. A hidden panel can be restored through the same menu.

Central tabs are Trajectory 2D, 3D View, Time Series and Samples. Choose the tab that matches the analysis you need.

The status bar reports ready, running, completed or failed. Docks can be moved or hidden; their layout is saved through Qt settings.

Control / action	Description
Left	Input parameters.
Centre	Plots and samples.
Right	Summary results.
Bottom	Event journal.

Practical note
If a panel disappears, do not reinstall the application; open View first and restore the required dock.

File menu and toolbar

The File menu controls the experiment data lifecycle. The toolbar duplicates core actions so frequent commands can be used without opening a menu.

Open Experiment opens a JSON picker. After validation, the specification loads and updates the left parameter panel.

Save Experiment writes the current setup. This is an input file and not necessarily the full numerical output.

JSON export stores the specification, trajectory and provenance. CSV export creates a sample table. PDF export creates a scientific report in the selected language.

Ctrl+O opens, Ctrl+S saves and F5 runs. Editing fields clears the old trajectory so it is not shown for new inputs.

Control / action	Description
Ctrl+O	Open JSON.
Ctrl+S	Save setup.
F5	Validate inputs and run.
Export	JSON, CSV or PDF after a successful run.

Practical note
For reproducibility, save the input JSON before a run and the result JSON after it.

Object parameters: mass and diameter

The Particle group sits at the top of the parameter dock. It defines the experiment name, mass and full diameter of the spherical object. Units are displayed beside numeric fields.

Mass must be finite and positive. The field supports small micro-particle values with extended decimal precision.

Diameter is the full sphere diameter, not radius. It is used for area, volume and Reynolds number. If you know radius, multiply it by two.

Particle density is not a separate field in the current form. Buoyancy correction uses particle mass and displaced volume derived from diameter.

For a non-spherical object, an equivalent diameter does not account for orientation-dependent drag. Such objects are outside the baseline application scope.

Control / action	Description
Mass	kg; strictly positive.
Diameter	m; full sphere diameter.
Name	Report label, not a physical parameter.

Practical note
If a value is rejected, check the field range and decimal separator. Pint validates dimensions at the input boundary.

Initial position and velocity

The Initial State group defines Cartesian coordinates and velocity components. x and z are horizontal; y is vertical and points upward.

Initial x, y and z are entered in metres. With ground stopping enabled, initial y cannot be below zero.

Velocity components vx, vy and vz are entered in m/s. The current form uses components rather than an angle field; resolve an oblique velocity into components.

Positive vy means upward motion; negative vy means downward motion. Default gravity_y is negative, so acceleration points along −y.

Wind is a separate vector. Drag uses relative velocity v_particle − v_wind, so wind direction affects resistance.

Control / action	Description
x/y/z	Position in metres.
vx/vy/vz	Velocity in m/s.
y sign	Positive direction is upward.

Practical note
Check coordinate signs with a simple vacuum example before enabling a complex drag model.

Maximum time and stopping conditions

The time field defines the upper bound of a run. It does not require the simulation to run for that entire duration: with the ground event enabled, the run may end earlier.

The output step defines saved sample spacing. If the duration is not a multiple of the step, the last interval is shortened so the run ends exactly at maximum time.

The stop-at-ground checkbox enables the first downward crossing of y=0. RK4 localizes the event within its step; DOP853 uses solve_ivp's event mechanism.

If unchecked, the mathematical path continues to maximum time even below y=0. This is an abstract simulation mode, not a physical model of motion through ground.

For DOP853 the output step is not the internal adaptive step. The solver chooses internal steps using rtol and atol. For RK4 the step controls integration and saved grid.

Control / action	Description
Maximum time	Upper time bound.
Output step	Spacing between saved samples.
Stop at y=0	First downward crossing of the surface.

Practical note
If you need to continue beyond the ground crossing, disable the event and state that assumption explicitly.

Environment: gravity and wind

The Environment group defines the gravity vector and a constant wind vector. All components are exposed because the physics core integrates a spatial state.

The default gravity_y near −9.80665 m/s² represents downward acceleration. Other gravity components are normally zero.

Wind is entered in m/s along x, y and z. A positive component means air moves along the positive axis direction.

The current form uses a constant wind vector along the trajectory. Height-dependent or layered wind requires a separate model and is not entered in the current UI.

With drag disabled, wind does not change the path because it only enters the relative-speed calculation for drag.

Control / action	Description
gravity_y	Negative means downward.
wind_x/y/z	Air velocity along three axes.
Profile	Wind is constant in the current form.

Practical note
Do not mix acceleration and velocity: these fields have different units and physical roles.

Atmosphere and its properties

The atmosphere module returns density ρ, temperature T, dynamic viscosity μ and speed of sound a. These properties are needed for drag force, Reynolds and Mach calculations.

ISA is a standard height profile. It uses geopotential height for layers and Sutherland's law for viscosity. ISA property inputs are disabled because the model defines them.

Constant atmosphere fixes density, temperature and dynamic viscosity. It is useful for laboratory conditions where the medium is assumed uniform.

Exponential atmosphere defines density through a scale height and temperature through a lapse rate. Viscosity follows temperature via Sutherland's law.

Dynamic viscosity is measured in Pa·s. Do not substitute kinematic viscosity in m²/s; it is a different quantity and would produce an incorrect Reynolds number.

Control / action	Description
ISA	Standard atmosphere.
Constant	Constant ρ, T and μ.
Exponential	Exponential density and temperature profile.

Practical note

For precision experiments, use measured medium properties when the standard atmosphere does not represent laboratory conditions.

Drag models and applicability

The built-in list includes Stokes, Schiller–Naumann, composite CGW/Clift–Gauvin, constant Cd and tabular Cd(Re). User-defined models are available through the manager.

Stokes is intended for very low Re and spherical laminar flow. Cd=24/Re can exceed 100 at small Re; this is valid for the built-in correlation.

Schiller–Naumann is a sphere correlation with Cd depending on Reynolds within its stated range. CGW/Clift–Gauvin combines documented correlation segments; observe applicability limits.

Constant Cd is a simplification for a limited regime. It does not model drag transitions and is not a universal curve.

A Cd(Re) table must contain increasing positive Re and positive Cd. Outside-range behaviour is selected as error, clamp or extrapolate.

Control / action	Description
Stokes / SN	Sphere drag correlations.
Constant Cd	Simplified constant model.
Table	Experimental Cd(Re) points.
Validity	Check limits and warnings.

Practical note
An applicability warning is important scientific information. Do not omit it from the final report.

Numerical methods and tolerances

RK4 and DOP853 solve the same equations but handle step size and error control differently. Compare them using physical quantities and tolerances.

RK4 evaluates the right-hand side four times per step. Height, atmosphere, relative speed, Reynolds and Cd are refreshed at each stage.

DOP853 uses adaptive SciPy integration. rtol is the relative tolerance, atol is absolute; a vector atol may be used for state components.

For regression control, fix RK4 and the step. For an independent check, compare with DOP853 and record both tolerances and the SciPy version.

A visually smooth line does not prove convergence. Check apogee, event time, displacement, final speed and changes under smaller steps.

Control / action	Description
RK4	Fixed step.
DOP853	Adaptive steps.
rtol/atol	Control local numerical error.

Practical note
The DOP853 output step does not control its internal adaptive step. Change rtol/atol to control accuracy.

Running and validating a calculation

The RUN button collects form values, validates them through Pydantic and Pint, converts quantities to SI and starts the calculation in a separate worker thread.

If a required value is non-finite/non-positive or dimensionally inconsistent, the run does not start. The validation message identifies the field or condition to fix.

A separate QThread prevents numerical work from blocking the main window. A second run is disabled while the current task is active.

Drag applicability warnings are written to the journal. A warning may accompany a successful run, but it must be considered when interpreting the result.

Editing any input clears the old result. This prevents a trajectory from previous parameters being mistaken for the current setup.

Control / action	Description
Validation	Pydantic + Pint.
Execution	Separate calculation thread.
After edits	Previous trajectory is cleared.

Practical note
Save the experiment JSON before a long series of runs.

Trajectory 2D

The 2D tab displays the x-y projection: x horizontally and y vertically. It is the primary view for height and horizontal displacement analysis.

A vertical line with constant x and changing y is expected for pure vertical falling. It does not indicate a plotting defect.

Auto scaling fits the axes to the data. If lateral displacement is very small, the path will look almost vertical.

With ground stopping enabled, the last point is y=0. With stopping disabled, the calculation may continue below zero until maximum time.

For an oblique path, set nonzero vx and vy. Without drag and with constant gravity, the path is parabolic.

Control / action	Description
x	Horizontal coordinate.
y	Height, upward axis.
Vertical line	Expected without lateral motion.

Practical note
Use the Samples tab for exact coordinates rather than estimating from screen pixels.

3D View

The 3D tab displays the spatial path in x, y and z. It is useful for lateral displacement analysis but depends on OpenGL and the graphics driver.

Rotate the scene with the mouse to inspect the path from different directions. The wheel usually changes scale; exact behaviour depends on the OpenGL widget.

Compare the scene with numeric x/y/z in the table. If z is constant, the path should remain in the x-y plane.

If OpenGL is unavailable, check 2D and the sample table. A visualization error does not mean the physics core failed.

For diagnosis, provide the GPU model, driver version and journal message. Try a local Windows session if the application is running through Remote Desktop.

Control / action	Description
x-y	Vertical plane.
z	Lateral displacement.
OpenGL	Depends on GPU driver.

Practical note
The 3D scene visualizes an already calculated path; it is not a separate solver.

Time-series plots

The Time Series tab displays speed, Mach number and kinetic energy. All quantities belong to the same result and share the same time axis.

Speed is particle speed magnitude. It differs from air-relative speed used in drag when wind is present.

Mach is flow-relative speed divided by local speed of sound. Atmospheric temperature and height may change sound speed.

Kinetic energy is 1/2 m v^2 and uses particle speed. For constant mass, energy follows the square of speed.

For a drop from rest, speed normally increases. With drag, the increase may slow as terminal speed is approached if that regime is reached within the run.

Control / action	Description
Speed	Particle speed magnitude, m/s.
Mach	Relative speed / speed of sound.
Energy	Kinetic energy, J.

Practical note
With wind, do not confuse particle speed and air-relative speed; they are used in different quantities.

Samples table

The table displays saved trajectory time samples: coordinates, velocity components, speed magnitude, Mach, energy, Reynolds and Cd.

RK4 generally outputs fixed-step points; the last step may be shortened. DOP853 returns the requested grid and appends a localized event point when stopping is triggered.

Monospaced text and alignment help compare values. Headers include units. Sorting is disabled to preserve time order.

At zero relative speed, initial Reynolds and Cd may be zero diagnostic values because drag is not evaluated at zero flow.

Use CSV export for Excel, Python or R; it preserves order and numeric precision better than manual copying.

Control / action	Description
t	Time in seconds.
x/y/z	Position in metres.
v	Velocity in m/s.
Re/Cd	Dimensionless indicators.

Practical note
Do not sort by an arbitrary column when analysing the chronological path.

Results dock

The right dock displays a summary: displacement, elapsed time, ground event, maximum height, apogee time, speed, Mach, Reynolds, energy and checksum.

Actual elapsed time may be shorter than maximum time when the ground event fires. If no event occurs, the UI reports that it was not reached.

Maximum height is interpolated between neighbouring samples using position and vertical velocity. The apogee may therefore lie between table rows.

Maximum speed, Mach and Reynolds are evaluated over saved points. Final energy belongs to the last trajectory state.

The checksum is based on quantized columns for regression detection. It does not promise bitwise identity across all CPUs.

Control / action	Description
Apogee	Interpolated maximum y.
Impact	First y=0 event time.
Checksum	Quantized regression hash.

Practical note
If a value looks unexpected, inspect the inputs, units and corresponding samples.

Saving and opening experiments

An experiment JSON is a portable description of the problem setup. It contains dimensional inputs with units, atmosphere, drag model, method and integrator settings.

Save the experiment through File → Save. Use a descriptive name that does not contain personal information if the file will be shared.

Open it through File → Open. Pydantic validates the schema and Pint checks units and converts them to SI. Unknown fields and invalid dimensions should fail validation.

A tabular model can be embedded in JSON. A plugin:<id> requires the corresponding user model on the receiving computer; review Python source separately.

After loading, review mass, diameter, initial state, atmosphere, drag, method and duration before running.

Control / action	Description
Save	Input setup.
Validation	Schema and physical dimensions.
Before run	Review parameters visually.

Practical note
Successful import means the format passed validation; it does not guarantee scientific applicability.

JSON and CSV export

Result JSON stores the setup, trajectory and reproducibility metadata. CSV is convenient for external processing of numeric samples.

JSON includes application, Python, NumPy, SciPy and Pint versions, platform, integrator, step or tolerances, seed, model description, model SHA-256 and trajectory SHA-256.

CSV contains time series and derived quantities. When opening it in Excel, check delimiter and decimal separator because locale settings may change import behaviour.

A checksum can reveal a regression but does not identify which quantity changed. Use separate physical tolerances for comparison.

Before public sharing, inspect experiment name, metadata, comments and filename for accidental personal information.

Control / action	Description
JSON	Setup + result + provenance.
CSV	Tabular samples.
Sharing	Check metadata and filenames.

Practical note
Do not edit a checksum manually; create a new result after changing data.

PDF report and language selection

The PDF report combines experiment parameters, model, applicability, results, numerical method, provenance, warnings and plots.

Choose File → Export Scientific PDF Report, filename and language: Russian, English or bilingual. A Unicode font is required for Cyrillic.

Check title, parameter table, drag-model metadata, step and tolerances, seed, library versions and both control hashes.

The bilingual report is longer and suitable for inter-laboratory exchange. Language modes share the same scientific data; only headings and descriptions are translated.

Keep the PDF together with input JSON and result JSON when a reproducible package is required.

Control / action	Description
Russian	Report with Russian headings.
English	Report with English headings.
RU + EN	Bilingual document.

Practical note
A PDF documents the calculation; it does not certify the selected physical correlation.

Drag Model Manager

The manager is available from Experiment → Drag Model Manager. It lists built-in and user models, imports tables and Python files and refreshes the registry.

User models are stored in the user's .trajectory_lab/models directory. Built-in models ship with the application and should not be removed as user models.

Before selection, check name, version, author, source, geometry and Reynolds range. The source should be traceable.

JSON/CSV tables are parsed as data and do not execute Python. Prefer them for simple Cd(Re) relationships.

Changing the model invalidates the previous run. Recalculate and save updated provenance.

Control / action	Description
Refresh	Read the model directory.
Import table	Add JSON/CSV.
Import Python	Copy source; execution requires confirmation.

Practical note
A model appearing in the manager does not mean it is valid for a particular experiment range.

Tabular Cd(Re) models

A tabular model stores Reynolds/Cd pairs. Reynolds must be positive and strictly increasing; Cd must be finite and positive.

Import a CSV with Reynolds and Cd columns or fill the editor manually. Check headers, delimiter, decimal separator and the edge points.

Linear interpolation connects points in ordinary coordinates. Log-log interpolates in logarithmic Re and Cd and suits power-law sections. Cubic log-log may smooth a curve but oscillate on sparse data.

Error stops the run outside the table. Clamp holds the nearest endpoint. Extrapolate extends the curve beyond measured data and requires justification.

Record geometry, source, author and version. Do not present a demonstration table as experimental data.

Control / action	Description
Linear	Ordinary linear interpolation.
Log-log	Interpolation in logarithmic coordinates.
Outside	Error / clamp / extrapolate.

Practical note
Do not extrapolate far beyond the measured range without independent validation.

Python plugins and security

A Python plugin is executable code. BaseDragModel defines the expected methods, but it does not provide security isolation. Code runs with the current Windows user's permissions.

Open source in an editor and inspect imports and side effects. Pay attention to subprocess, os, socket, urllib, requests, ctypes, winreg and dynamic execution.

Do not download unknown plugins from forums. Do not run the application as Administrator for plugin use. Python code may read and modify files, start processes and access the network.

The CLI does not load Python plugins by default; the explicit --allow-python-plugins flag is required. The GUI asks for confirmation before execution.

If a model can be represented by points, use JSON/CSV: a declarative table does not execute arbitrary Python.

Control / action	Description
JSON/CSV	Preferred data format.
Python	Source review required.
Sandbox	No full isolation is provided.

Practical note
A UI warning does not make untrusted Python code safe.

Language, themes and docks

The View menu controls language, theme and dock visibility. RU/EN switches without restart through translation catalogs; themes are applied through QSS.

View → Language switches Russian and English. The catalog should update menu headings, docks, tabs, table and messages.

View → Theme offers Light, Dark and Soft Pink. Pink uses a pastel accent while retaining neutral panels and a technical style.

Parameters, Results and Journal docks can be hidden or restored through View. Window size and state are saved in QSettings.

Do not delete user settings during an update if you want to retain language, theme and layout.

Control / action	Description
Language	RU/EN without restart.
Theme	Light/Dark/Soft Pink.
Docks	Show or hide panels.

Practical note
Changing the theme does not modify experiment inputs or results.

Local data and privacy

The main application operates locally. Settings, user models and logs are stored in the current user's profile; the project has no built-in telemetry or experiment-file upload.

User data are normally stored in %USERPROFILE%\.trajectory_lab. It may contain models, logs and settings. Do not publish the whole directory.

The local log filter replaces known user path roots but is not a universal personal-data scrubber. Review logs manually before sharing.

Builder contacts python.org and PyPI when installing the runtime and libraries. This network activity belongs to initial setup, not to calculations in the desktop application.

Python plugins can access the network and transmit data independently. Load only reviewed code. SHA-256 proves file identity, not benign behaviour.

Control / action	Description
Local	Settings, models, logs and saved experiments.
Builder	python.org and PyPI during first build.
Plugins	May perform arbitrary network actions.

Practical note
Before publishing JSON, CSV, PDF or logs, remove names, paths and values that should remain private.

Reproducibility and checksums

Result provenance records the application, Python, NumPy, SciPy and Pint versions, OS, integrator, step, tolerances, seed, model and its SHA-256.

For RK4, fix the step and dependency versions. For DOP853, store rtol and atol. Compare individual physical quantities against defined tolerances.

The trajectory checksum uses per-column normalization and quantization. It helps detect regressions but does not promise bitwise identity across CPUs.

Seed is stored for audit and future stochastic extensions. It may not affect deterministic examples.

For scientific exchange, keep input JSON, result JSON, CSV and PDF with the environment version description.

Control / action	Description
Input	JSON and model parameters.
Numerical control	Step or rtol/atol.
Check	Per-quantity tolerances + checksum.

Practical note
A checksum mismatch alone does not explain the cause; compare values and software environment versions.

Vacuum reference case

The vacuum_reference.json example disables drag and uses constant gravity. It is a reference case with an analytical solution.

Open the example and check position, velocity, g, maximum time, step and method before running.

For vertical position use $y(t)=y0+vy0 \cdot t+0.5 \cdot g_y \cdot t^2$. With no horizontal forces, $x(t)=x0+vx0 \cdot t$.

If vx=0, the x-y graph is vertical. If vy is positive, the object first rises; after apogee, vertical velocity changes sign.

Compare apogee time, maximum height, final speed and event time with analytical values within numerical tolerance.

Control / action	Description
Purpose	Check coordinates and integrator.
Drag	Disabled.
Plot	A vertical line is possible when vx=0.

Practical note
Do not interpret drag results until the vacuum reference agrees with the analytical solution.

Air-resistance example

The air_resistance_drop.json example uses ISA and Schiller–Naumann for a falling sphere. It checks the atmosphere → viscosity → Reynolds → Cd → acceleration chain.

Confirm ISA and Schiller–Naumann; review mass, diameter, initial height and output step.

Speed should rise from rest. With drag, its growth may slow as terminal speed is approached; a finite interval may be too short to reach it.

Re=ρ·v_rel·d/μ uses diameter. Mach is based on local speed of sound. Check the table and time-series plots.

If the journal reports an applicability warning, include it in the interpretation and do not compare the model beyond its stated conditions.

Control / action	Description
Speed	Changes over time.
Re	Depends on ρ, v, d and μ.
Cd	Depends on the selected correlation.

Practical note
Demonstration inputs do not replace calibration of your own laboratory setup.

Journal and troubleshooting

The journal records application and calculation messages. To diagnose a problem, keep the messages leading up to the failure, not only the final line.

For a calculation failure, save JSON, method, atmosphere, drag model and journal. Include application version and minimal reproduction steps.

For a 3D issue, provide GPU and driver. If the table and 2D view contain data, the issue may be limited to OpenGL rendering.

For a Builder failure, save the complete stage log, Windows version, network status and exact pip/PyInstaller message.

Before sharing logs, manually remove personal paths, filenames and research data you do not want to disclose.

Control / action	Description
Calculation	JSON + journal + inputs.
3D	GPU and driver.
Builder	Full build log.

Practical note
Do not send the entire .trajectory_lab folder; it may contain user models and experiments.

Updates and backups

The application folder and user data are separate. During an update, replace the application directory while retaining .trajectory_lab in the user profile.

Close the application and back up the user data directory. A typical Windows path is %USERPROFILE%\.trajectory_lab.

Extract the new version separately and test it before deleting the previous one. Do not overwrite a running application.

Transfer only required user models. Review .py source before copying; JSON/CSV tables are easier and safer to migrate.

Delete the old folder only after verifying old JSON loading, calculation and export in the new version.

Control / action	Description
Backup	Copy .trajectory_lab.
Update	Extract the application separately.
Cleanup	Remove the old folder after validation.

Practical note
Do not delete the user data folder during an update; it may contain your models and settings.

Final acceptance checklist

Before accepting a result, check more than a visually pleasing plot: verify the setup, dimensions, model applicability and convergence.

Review mass, diameter, position, velocity, gravity, wind, atmosphere, drag model and maximum time.

Confirm units and account for applicability warnings. For RK4 test step sensitivity; for DOP853 record rtol and atol.

Inspect results dock, plots and table. Compare numerical values for key quantities, not only visual shape.

Save input JSON, result JSON, CSV and PDF. Before sharing, ensure files contain no private paths or names.

Control / action	Description
Science	Reference case and model validity.
Numbers	Tolerances and convergence.
Records	JSON, CSV, PDF and sanitized journal.

Practical note
If model applicability is uncertain, state it in the report and do not present the result as a universal prediction.